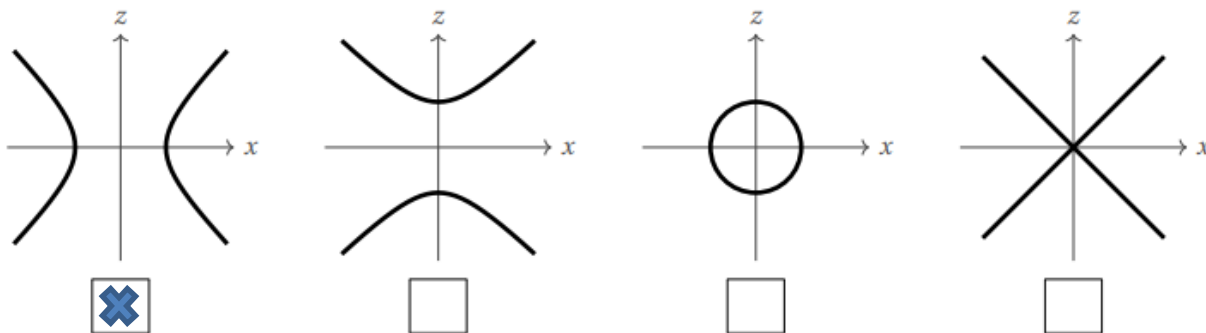


MTH 202 Chapter 1: Three-Dimensional Space & Vectors

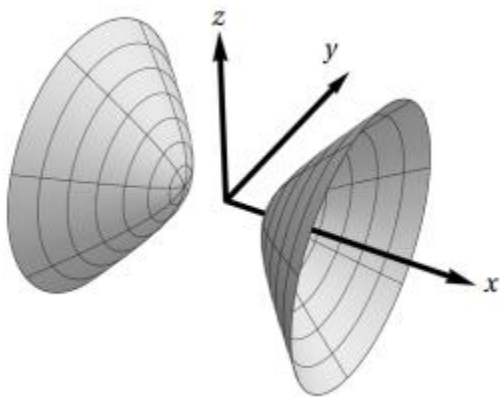
1.7 Quadric Surfaces Worksheet

1. Consider the quadric surface Q defined by the equation $x^2 + y^2 - z^2 = 1$. Check the box below the picture of the curve formed by intersecting Q with the xz -plane.



If Q intersects the xz -plane, then this means that at the intersection $y = 0$ and this means that our equation becomes $x^2 - z^2 = 1$ which is a hyperbola open down the negative and positive x -axis.

2. Consider the hyperboloid H of two sheets shown below. Circle the equation of H .



$y^2 + z^2 - x^2 = 1$	$x = y^2 + z^2 + 1$
$x^2 + y^2 + z^2 = -1$	<u>$x^2 - y^2 - z^2 = 1$</u>

We see that when x is a constant, we see our equation should generate circles of vary radii. When $x < a$ (for some value of a) we also see that no circles exist.

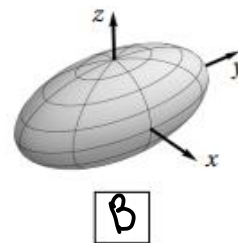
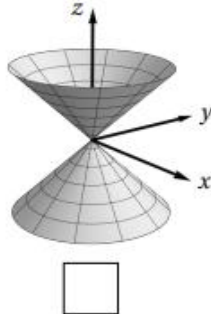
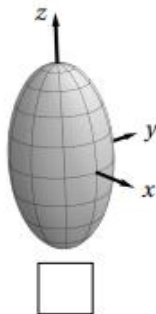
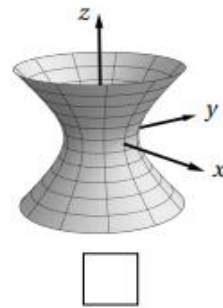
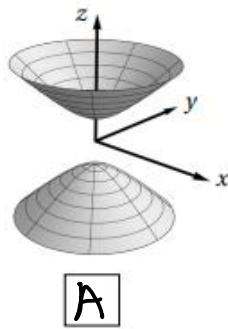
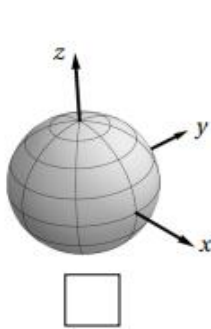
When y or z are constant, we see our equations should generate hyperbolas.

The only equation that accomplishes this is $x^2 - y^2 - z^2 = 1$.

3. For each equation below, write the corresponding letter in the box next to the picture of the surface it describes.

(A) $x^2 + y^2 - z^2 + 1 = 0$

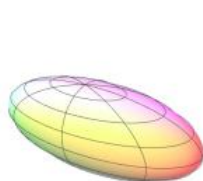
(B) $4x^2 + y^2 + 4z^2 - 1 = 0$



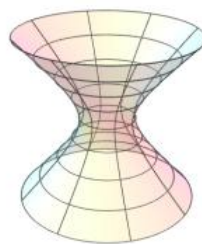
(A) If we let x or y equal a constant we see that the equation will generate hyperbolas. If z is constant (and greater than 1) the equation will generate circles. However, if $z < 1$ then there are no solutions to the equation. The surface that matches this description is the top, middle surface.

(B) If x , y , or z are constant then the equation will generate ellipses. However, more specifically, we see that if y is a constant the equation will generate circles. The surface that matches this description is the bottom, right surface.

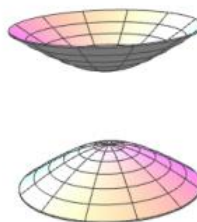
4. Let $f(x, y, z) = ax^2 + by^2 + cz^2$ for some real numbers a , b , and c . Which of the following *could not* be a level set of f ?



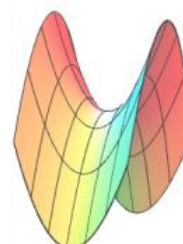
A. Ellipsoid



B. Hyperboloid of 1 sheet



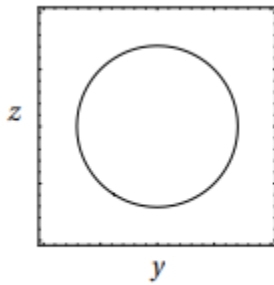
C. Hyperboloid of 2 sheets



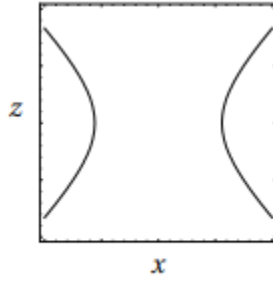
D. Hyperbolic paraboloid

We notice that if $d = ax^2 + by^2 + cz^2$ then if either x , y , or z are constant then this equation will correspond to either an ellipse or a hyperbola (or have no solutions for appropriately selected values). As such, the only surface which is *not* possible is the hyperbolic paraboloid since this equation will never generate parabolas. Therefore, we select D.

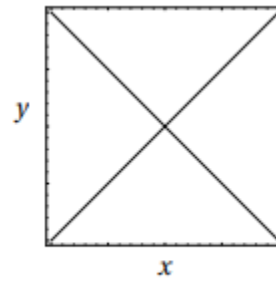
5. Suppose S is a quadric surface whose intersections with the following three planes are:



Plane $\{x = 1\}$

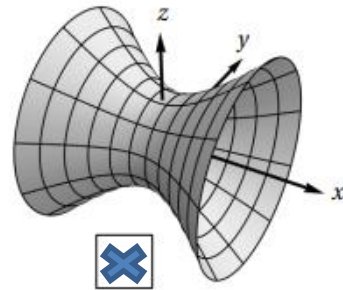
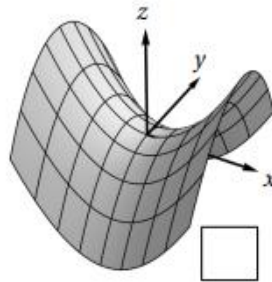
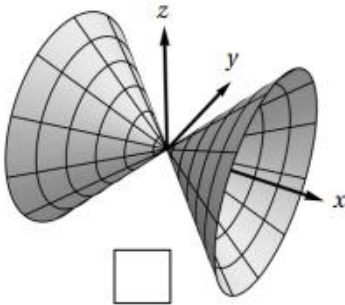


Plane $\{y = -1.5\}$



Plane $\{z = 1\}$

- a) Mark the box next to the picture of the portion of S where $-2 \leq x \leq 2$, $-2 \leq y \leq 2$, and $-2 \leq z \leq 2$.



- b) Circle the equation that S satisfies:

$$x^2 + y^2 - z = 0$$

$$y^2 + z^2 - x^2 = 1$$

$$y^2 + z^2 = x^2$$

We see that if x is constant then our equation should produce only circles of varying radii.
If y or z are constant, then our equation should produce hyperbolas.
The only equation that satisfies these conditions is the middle one.

6. Determine quadric surface is represented by each of the following equations.

a) $x^2 + y + 2z^2 = 1$

If we let y equal a constant we see that the equation will generate ellipses. If x or z is constant the equation will generate parabolas. Therefore, this equation describes an elliptical paraboloid.

b) $10x^2 + 4y^2 - z^2 = 0$

If we let z equal a constant, we see that the equation will generate ellipses (assuming that $z \neq 0$). If $z = 0$ then the equation will generate a single point. If x or y is constant the equation will generate hyperbolas. Therefore, this equation describes a cone.

c) $\frac{1}{2}x + y^2 - z^2 = 2$

If we let x equal a constant, we see that the equation will generate hyperbolas. (assuming that $z \neq 0$). If we let y or z equal a constant then the equation will generate parabolas. Therefore, this equation describes a hyperbolic paraboloid.